# Investigación Operativa

Dualidad

# Teorema Fundamental de la Dualidad

Dado un programa lineal en su forma canónica y su correspondiente programa dual asociado,

| Problema original | | Problema dual |
|---|---|---|
| PRIMAL | | |
| $\text{Max } z = cx$ | | $\text{Min } w = b^T u$ |
| s.a | | s.a |
| $Ax \leq b$ | | $-A^T u \geq c^T$ |
| $x \geq 0$ | | $u \geq 0$ |

Max $z = 20x_1 + 45x_2$
s.a $x_1 + 2x_2 \leq 40$
$3x_1 + 1.5x_2 \leq 75$
$x_2 \leq 15$
$x_i \geq 0$

Min $w = 40u_1 + 75u_2 + 15u_3$
s.a $u_1 + 3u_2 \geqslant 20$
$2u_1 + 1.5u_2 + u_3 \geqslant 45$
$u_j \geqslant 0$

El teorema enuncia que

> Si el programa primal tiene solución óptima finita, los valores de $u$ que minimizan el dual son los costos marginales del programa original. Además, el valor óptimo de la función objetivo del dual coincide con el valor óptimo de la función objetivo del original $(w^* = z^*)$.

$U^{*T} = C_B B^{-1} = Z_j$

$w^* = z^*$

# En la tabla

| $c_i$ | $A_i$ | 20 | 45 | 0 | 0 | 0 | $x_i$ | $x_i/y_{ij}$ ($y_{ij}>0$) |
|---|---|---|---|---|---|---|---|---|
| | | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $A_5$ | | |
| 20 | $A_1$ | 1 | 0 | 1 | 0 | –2 | 10 | |
| 0 | $A_4$ | 0 | 0 | –3 | 1 | 9/2 | 45/2 | |
| 45 | $A_2$ | 0 | 1 | 0 | 0 | 1 | 15 | |
| $z_i$ | | 20 | 45 | 20 | 0 | 5 | $z = 875$ | |
| $c_i - z_i$ | | 0 | 0 | –20 | 0 | –5 | | |

Costo marginal o Valor implícito

$$c_B B^{*-1} = z_j$$

# Relación de *x* con *u*

Problema original

$$\text{Max } z = 20x_1 + 45x_2$$

s. a

$$\begin{aligned} x_1 + 2x_2 &\le 40 \\ 3x_1 + 1{,}5x_2 &\le 75 \\ x_2 &\le 15 \\ x_1, x_2 &\ge 0 \end{aligned}$$

| Tipo | Vector x | Coef x | Vector u | Coef u | Tipo |
|---|---|---|---|---|---|
| Concreta | $x_1$ | 20 | $u_4$ | 0 | Exceso |
| Concreta | $x_2$ | 45 | $u_5$ | 0 | Exceso |
| Holgura | $x_3$ | 0 | $u_1$ | 40 | Concreta |
| Holgura | $x_4$ | 0 | $u_2$ | 75 | Concreta |
| Holgura | $x_5$ | 0 | $u_3$ | 15 | Concreta |

Problema dual

$$\text{Min } w = 40u_1 + 75u_2 + 15u_3$$

s. a

$$\begin{aligned} u_1 + 3u_2 &\ge 20 \quad (u_4) \\ 2u_1 + 1{,}5u_2 + u_3 &\ge 45 \quad (u_5) \\ u_1, u_2, u_3 &\ge 0 \end{aligned}$$

# Valores del Primal en el Dual y viceversa

Problema original

Max $z = 20x_1 + 45x_2$

s. a

$$\begin{aligned} x_1 + 2x_2 &\le 40 \\ 3x_1 + 1{,}5x_2 &\le 75 \\ x_2 &\le 15 \\ x_1, x_2 &\ge 0 \end{aligned}$$

| $c_i$ | $A_i$ | 20 $A_1$ | 45 $A_2$ | 0 $A_3$ | 0 $A_4$ | 0 $A_5$ | $x_i$ | $x_i/y_{ij}$ ($y_{ij}>0$) |
|---|---|---|---|---|---|---|---|---|
| 20 | $A_1$ | 1 | 0 | 1 | 0 | –2 | 10 | |
| 0 | $A_4$ | 0 | 0 | –3 | 1 | 9/2 | 45/2 | |
| 45 | $A_2$ | 0 | 1 | 0 | 0 | 1 | 15 | |
| $z_j$ | | 20 | 45 | 20 | 0 | 5 | z = 875 | |
| $c_j - z_j$ | | 0 | 0 | –20 | 0 | –5 | | |

x*=(10, 15, 0, 22.5, 0). Base A1, A4, A2

| | |
|---|---|
| $x_1$ | $u_4$ |
| $x_2$ | $u_5$ |
| $x_3$ | $u_1$ |
| $x_4$ | $u_2$ |
| $x_5$ | $u_3$ |

u*=(20, 0, 5, 0, 0). Base A1, A3

Problema dual

Min $w = 40u_1 + 75u_2 + 15u_3$

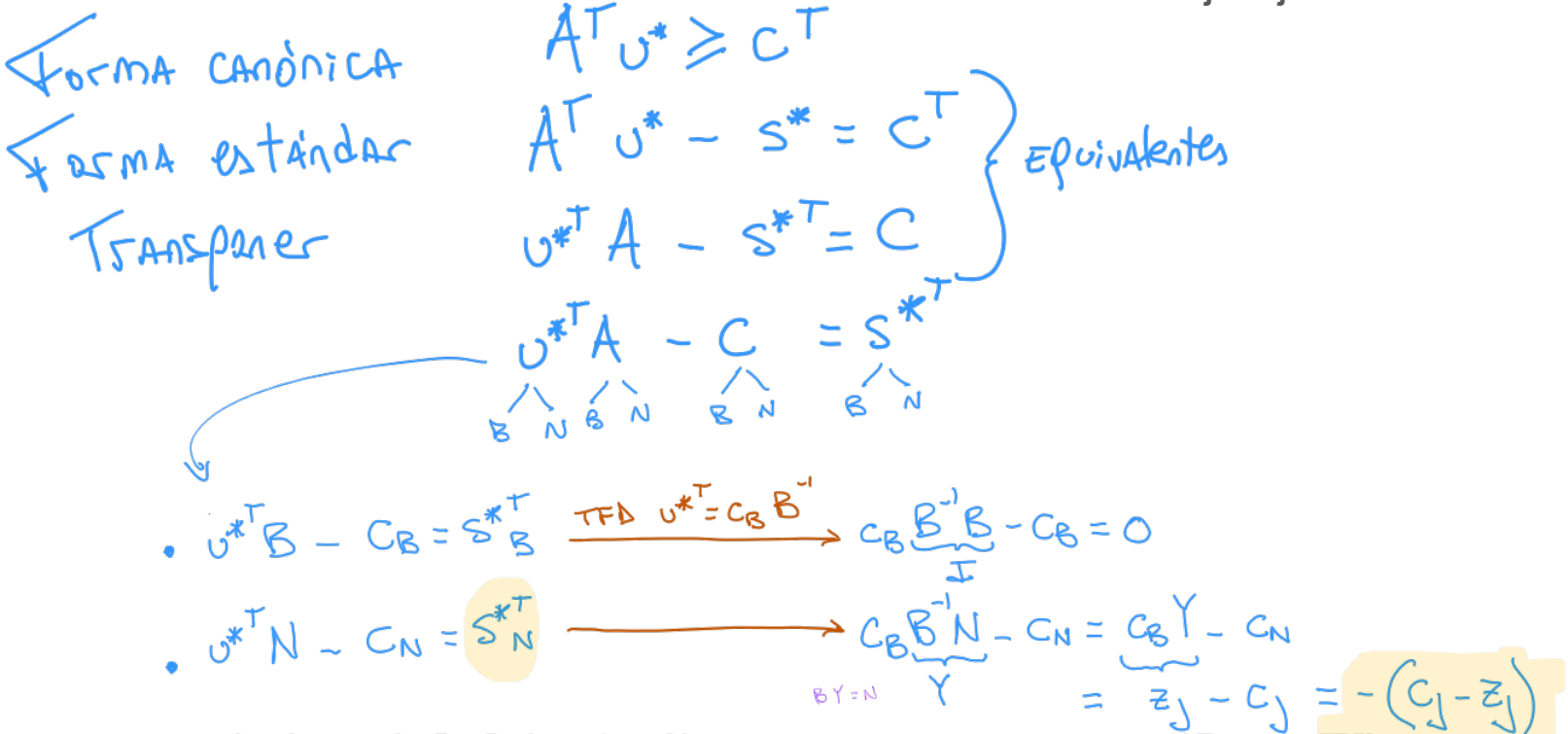
s. a

$$\begin{aligned} u_1 + 3u_2 &\ge 20 \\ 2u_1 + 1{,}5u_2 + u_3 &\ge 45 \\ u_1, u_2, u_3 &\ge 0 \end{aligned}$$

| $c_i$ | $A_i$ | 40 $A_1$ | 75 $A_2$ | 15 $A_3$ | 0 $A_4$ | 0 $A_5$ | $x_i$ | $x_i/y_{ij}$ ($y_{ij}>0$) |
|---|---|---|---|---|---|---|---|---|
| 40 | $A_1$ | 1 | 3 | 0 | –1 | 0 | 20 | |
| 15 | $A_3$ | 0 | –9/2 | 1 | 2 | –1 | 5 | |
| $z_j$ | | 40 | 105/2 | 15 | –10 | –15 | w = 875 | |
| $c_j - z_j$ | | 0 | 22,5 | 0 | 10 | 15 | | |

# Para los valores no base

Reemplazamos costo marginal por **costo reducido -($c_j$ - $z_j$)**



# Resumen

Problema original

$$\text{Max } z = 20x_1 + 45x_2$$

s. a

$$\begin{aligned} x_1 + 2x_2 &\le 40 \\ 3x_1 + 1{,}5x_2 &\le 75 \\ x_2 &\le 15 \\ x_1, x_2 &\ge 0 \end{aligned}$$

| $c_i$ | $A_i$ | 20 $A_1$ | 45 $A_2$ | 0 $A_3$ | 0 $A_4$ | 0 $A_5$ | $x_i$ | $x_i/y_{ij}$ $(y_{ij}>0)$ |
|---|---|---|---|---|---|---|---|---|
| 20 | $A_1$ | 1 | 0 | 1 | 0 | –2 | 10 | |
| 0 | $A_4$ | 0 | 0 | –3 | 1 | 9/2 | 45/2 | |
| 45 | $A_2$ | 0 | 1 | 0 | 0 | 1 | 15 | |
| $z_j$ | | 20 | 45 | 20 | 0 | 5 | $z = 875$ | |
| $c_j - z_j$ | | 0 | 0 | –20 | 0 | –5 | | |

x*=(10, 15, 0, 22.5, 0).
Base A1, A4, A2

| | | Primal | Dual | Dual | Primal |
|---|---|---|---|---|---|
| $x_1$ | $u_4$ | $x_1 = 10$ | $-z_4 = 10$ | $u_1 = 20$ | $z_3 = 20$ |
| $x_2$ | $u_5$ | $x_2 = 15$ | $-z_5 = 15$ | $u_2 = 0$ | $z_4 = 0$ |
| $x_3$ | $u_1$ | $x_3 = 0$ | $c_1-z_1 = 0$ | $u_3 = 5$ | $z_5 = 5$ |
| $x_4$ | $u_2$ | $x_4 = 22.5$ | $c_2-z_2 = 22.5$ | $u_4 = 0$ | $-(c_1-z_1) = 0$ |
| $x_5$ | $u_3$ | $x_5 = 0$ | $c_3-z_3 = 0$ | $u_5 = 0$ | $-(c_2-z_2) = 0$ |

u*=(20, 0, 5, 0, 0).
Base A1, A3

Problema dual

$$\text{Min } w = 40u_1 + 75u_2 + 15u_3$$

s. a

$$\begin{aligned} u_1 + 3u_2 &\ge 20 \\ 2u_1 + 1{,}5u_2 + u_3 &\ge 45 \\ u_1, u_2, u_3 &\ge 0 \end{aligned}$$

| $c_i$ | $A_i$ | 40 $A_1$ | 75 $A_2$ | 15 $A_3$ | 0 $A_4$ | 0 $A_5$ | $x_i$ | $x_i/y_{ij}$ $(y_{ij}>0)$ |
|---|---|---|---|---|---|---|---|---|
| 40 | $A_1$ | 1 | 3 | 0 | –1 | 0 | 20 | |
| 15 | $A_3$ | 0 | –9/2 | 1 | 2 | –1 | 5 | |
| $z_j$ | | 40 | 105/2 | 15 | –10 | –15 | $w = 875$ | |
| $c_j - z_j$ | | 0 | 22,5 | 0 | 10 | 15 | | |

DUALIDAD